

Ezra Ornstein

(647) 544-6662 | eorNSTEi@uwaterloo.ca | [linkedin.com/in/ezra-ornstein](https://www.linkedin.com/in/ezra-ornstein)

ENGINEERING PROJECTS

Exoskeleton Team Member – Waterloo Biomechatronics

Nov 2025 – Present

- Designed power electronics and power housing unit for Waterloo's first exoskeleton in the Applied Collegiate Exoskeleton Competition
- Engineered and validated PCBs by designing and simulating circuits in **Altium** and **LTSpice**
- Integrated removable batteries, battery management and cooling systems, and circuit boards into a compact unit

Mechanical Lead – Automated Air Hockey Game Defense Machine

Oct 2025 – Dec 2025

- Built autonomous air-hockey defense system to return incoming pucks, achieving **76%** successful deflection rate
- Adapted design to strict component constraints (VEX IQ parts only)
- Designed custom two-axis positioning system using gear-reduced belt drive for lateral motion and motor-driven linear actuator for rapid striking action
- Collaborated with software team on C++ control integration for real-time tracking and response

Discrete Transistor Fuzz Face Guitar Pedal

Mar 2025

- Designed and built analog distortion pedal using discrete NPN BJTs, adjusting resistor values to set transistor bias points and achieve stable amplification and clipping behavior
- Iteratively refined circuit behavior through empirical testing and component-level modifications
- Calculated correct resistor values and modified a schematic to fit available components

WORK AND VOLUNTEER EXPERIENCE

Events Team Member at Humber Polytechnic IT Services

Feb 2025 – Jun 2025

Humber Polytechnic

- Set up and troubleshoot integrated AV systems for internal and external events, achieving **98%** client satisfaction
- Diagnosed and resolved technical issues under time constraints, developing problem-solving skills in live production environments and ensuring seamless event execution
- Collaborated with event coordinators and technical staff to interpret system requirements and deliver reliable AV configurations for diverse venue configurations

Lifeguard at City of Toronto

Jan 2024 – Jun 2025

Douglas Snow Aquatic Centre

- Maintained vigilant surveillance of pools to ensure safety and prevent accidents in high-traffic environments
- Demonstrated ability to remain calm and make quick decisions under pressure by responding to emergency situations

President at Change Impacting Toronto's Youth

Sep 2024 – Jun 2025

John Polanyi Collegiate Institute

- Led team of 10 students coordinating community service initiatives including homelessness support, food security programs, clothing drives, and international humanitarian fundraising
- Organized school-wide engagement campaigns such as food drives and fundraising events to mobilize the broader community, raising thousands of dollars for local food banks and charities
- Projects included Impact 2024, partnered with Kids Against Hunger, where volunteers created and packaged 8000 freeze-dried meal bags to feed the hungry in Haiti

SKILLS AND INTERESTS

Tools: SOLIDWORKS, AutoCAD, Altium, LTSpice, Microsoft Suite, Google Suite

Languages: C++

Hobbies: Guitar, Rock Climbing, Swimming, Axe Restoration, Dune

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Mechatronics Engineering

Began Sep 2025

John Polanyi Collegiate Institute

Toronto, ON

Graduating Average: 99%

Sep 2021 – Jun 2025

Governor General's Award, Schulich Leader Nominee, Principal's Award for Student Leadership